

Q.21 Write the problems associated with hydrogen energy.

Q.22 Explain tidal energy and applications of tidal power plant.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 Describe the working of solar cell (photo-voltaic cell) with neat diagram. Also explain its V-I diagram and Solar cell efficiency.

Q.24 Write different types of wind mills. Explain with neat diagram the working of horizontal Axis wind turbine (HAWT)

Q.25 Explain fermentation and hydrolysis method of bio-mass conversion.

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6th Sem / Chemical (Pulp & Paper)

Subject :Alternate Energy Sources

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 Which of the following is green house gas?

- a) Methane b) nitrons oxide
- c) CO₂ d) Ethane

Q.2 Wind energy system converts the _____ energy of wind to mechanical power

- a) Heat b) kinetic
- c) potential d) None

Q.3 Wind energy _____ with increase of height

- a) increases
- b) decreases
- c) no change
- d) may increase or decrease

Q.4 Which two bands of Solar radiation are reaching earth maximum.

- a) UV & visible
- b) Visible and infrared
- c) Infrared & ultraviolet
- d) All

Q.5 What is the primary source of tidal energy?

- a) Solar radiation
- b) Wind energy
- c) Gravitational pull of sun & moon
- d) Geothermal energy

Q.6 What is the primary purpose of a fuel cell?

- a) To store electrical energy
- b) To convert chemical energy into electrical Energy
- c) To generate heat energy
- d) To produce mechanical energy

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 What is the non-conventional source of energy?

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Q.8 Name material used for making solar cell?

Q.9 Write two examples of primary source of energy.

Q.10 Define ocean energy.

Q.11 Write the full form of OTEC.

Q.12 Define tidal energy.

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 What are the non-conventional sources of energy? Also write its importance.

Q.14 Explain in brief the working of solar water heating system with neat diagram.

Q.15 Explain green house effect with neat diagram.

Q.16 Write the advantages and disadvantages of wind power plant.

Q.17 Define biomass and its composition.

Q.18 Explain in brief the wet process for conversion of bio-mass.

Q.19 Explain in brief the working of open OTEC power station.

Q.20 Define fuel cells and classify fuel cells.

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